The Geometry of Partisanship: Graph Attention Networks for Predicting Legislative Defection in the U.S. House, 1995–2024

Abstract

When a legislator breaks with their party, it reveals structural features of Congress that roll-call statistics alone cannot capture. We construct temporal voting networks for the U.S. House from the 104th through 118th Congress (1995–2024), documenting a 30-percentage-point decline in cross-party agreement and a collapse in the Fiedler value from 0.84 to 0.03, indicating near-complete partisan separation. Spectral decomposition confirms that the Fiedler vector correlates with DW-NOMINATE scores at $r > 0.97$, meaning the dominant axis of congressional ideology is recoverable from network topology alone. We introduce the Structural Resilience Index, finding that the Tea Party wave of 2010 produced an 93% decline in the system's capacity for structural recovery. We also explore whether graph neural networks can predict party-line defection from network structure. Both Graph Attention Networks and Graph Convolutional Networks achieve comparable performance (mean AUC 0.80–0.84), but are substantially outperformed by tree-based methods, suggesting that the extreme partisan structure of modern co-voting networks leaves little room for learned attention to improve over uniform aggregation.

1 Introduction

Start with a fact: in the 104th Congress (1995–96), roughly half of all House members voted against their party more than 10% of the time. By the 115th Congress (2017–18), that number had fallen to 6.7%. The disappearance of the party-line crosser is one of the defining features of contemporary American politics, yet the mechanisms that produce this sorting remain incompletely understood. Do members defect less because the parties have become more ideologically coherent, because institutional pressures have intensified, or because the kinds of people who run for Congress have changed? The answer is probably all three, but separating these channels requires a framework that captures the relational structure of legislative behavior, not just individual-level covariates.

This paper develops that framework. We model each session of the U.S. House as a graph in which nodes represent members and edges encode voting agreement, then deploy Graph Attention Networks to learn which structural features of the network predict who will break ranks. The intuition is straightforward: a legislator's propensity to defect depends not only on their own ideology and party, but on the voting behavior of the colleagues they are most connected to. A moderate Republican surrounded by hard-liners faces different incentive structures than one embedded in a bipartisan cluster. Standard regression cannot capture this relational context. Graph neural networks can.

The paper makes three contributions. First, we document the structural collapse of bipartisan connectivity over three decades using spectral graph theory. The Fiedler vector of the normalized Laplacian correlates with first-dimension DW-NOMINATE scores at $r > 0.97$, meaning the dominant axis of congressional ideology is recoverable from network topology alone. We introduce the Structural Resilience Index (SRI) to quantify the system's capacity for recovery from partisan fragmentation; it collapses by 93% after the 2010 Tea Party wave and remains near zero through the 118th Congress. We also develop the Bridge Legislator Index (BLI)

to identify members who occupy structurally central positions between parties, and conduct counterfactual perturbation experiments showing that just 5–10 members account for most bipartisan connectivity.

Second, we explore whether graph neural networks can predict party-line defection from network structure. Using a temporal train-test split (Congress 104–114 for training, 115–117 for testing), we find that tree-based methods substantially outperform graph approaches (random forests: 0.985 AUC vs. GAT: 0.802 AUC). We attribute this null result to the extreme structural simplicity of modern polarized networks: when same-party edges are structurally equivalent and cross-party edges uniformly uninformative, there is little room for learned attention to improve over uniform aggregation. The GAT's value lies not in classification accuracy but in the interpretable representations it produces—attention maps revealing 200:1 same-to-cross-party weight ratios, and embeddings enabling the BLI and related analyses.

Third, we use the temporal dimension of our data to examine critical junctures in recent congressional history. The Tea Party wave of 2010 drove cross-party agreement down from 0.52 (108th–111th) to 0.30 (112th–114th), a drop that persisted through the Trump era. Difference-in-differences analysis confirms this was a structural break, not a gradual trend. The McCarthy speaker vote in January 2023 provides a natural experiment: our model trained through the 117th Congress flagged 4 of 20 holdouts (20% recall, 11.8% precision), modestly exceeding the 8.7% base rate but illustrating the difficulty of predicting idiosyncratic political events.

2 Related Work

We position our contribution at the intersection of three literatures: congressional behavior and polarization, graph neural networks for relational data, and machine learning for political science.

2.1 Partisan Sorting and Polarization

The literature on partisan sorting emphasizes how American voters and elites have become increasingly aligned along party lines over the past several decades (Levendusky, 2009; Mason, 2018). This sorting has transformed congressional behavior, as party leaders gained leverage to enforce discipline and cross-party coalitions became rare (Poole and Rosenthal, 1985). Our work complements this literature by showing how network methods can quantify the structural consequences of sorting: the collapse of cross-party ties and the increasing dominance of intra-party connectivity.

2.2 Graph Neural Networks

Graph Attention Networks (Veličković et al., 2018) extend the message-passing framework of Graph Convolutional Networks (Kipf and Welling, 2017) by learning edge-specific attention weights. This anisotropic aggregation is particularly suited to legislative networks, where not all co-voting relationships carry equal informational weight. Our comparison between GAT and GCN isolates the value of learned attention: while the GCN's uniform aggregation performs comparably for classification, only the GAT reveals which specific edges drive predictions.

2.3 Machine Learning for Tabular Data

Recent work has documented that tree-based models often outperform deep learning on small, structured tabular datasets (Grinsztajn et al., 2022). Our results confirm this pattern: random forests achieve 0.980 mean AUC on congressional defection prediction, substantially exceeding the GAT's 0.816. Rather than viewing this as a failure of the neural approach, we treat it as a methodological boundary condition that clarifies the GAT's distinct contribution: interpretable relational structure rather than raw classification accuracy.

3 Data and Graph Construction

3.1 Voteview Roll-Call Data

We draw on the Voteview database (Lewis et al., 2023), which provides complete roll-call records and member-level metadata for every session of Congress. Our analysis spans the 104th through 118th Congresses (1995–2024), encompassing 15 sessions and over 9.7 million individual vote records. We restrict attention to House members affiliated with the two major parties (Democrats, coded 100; Republicans, coded 200), yielding between 438 and 455 members per congress.

For each session, we extract cast codes (yea, nay, and abstention variants) and collapse them into binary yea/nay indicators. Members who did not participate in a given roll call are excluded from that roll call's agreement computation. We also extract DW-NOMINATE scores, which serve as our primary ideological benchmark.

3.2 Agreement Networks

For each congress, we construct a pairwise agreement matrix $A \in \mathbb{R}^{n \times n}$, where A_{ij} records the fraction of roll calls in which members i and j voted the same way, conditional on both participating. We require a minimum of 10 shared roll calls for any pair; below this threshold, the agreement rate is set to zero. This yields dense, weighted adjacency matrices that encode the full structure of co-voting behavior.

To produce graphs suitable for message-passing neural networks, we sparsify each agreement matrix using k-nearest-neighbor construction with $k = 15$. Each member retains directed edges to the 15 colleagues with whom they agree most frequently. This approach preserves local voting structure while eliminating the computational burden of fully connected graphs (which contain roughly 100,000 edges per congress). The resulting graphs contain approximately 6,600 to 6,800 edges per session.

3.3 Node Features

Each member is represented by a six-dimensional feature vector:

1. **DW-NOMINATE Dimension 1**: The primary ideological coordinate (liberal–conservative).
2. **DW-NOMINATE Dimension 2**: The secondary dimension, historically associated with race and social issues.
3. **Party**: Binary indicator (Republican = 1, Democrat = 0).
4. **Seniority**: Number of recorded votes, normalized by 1,500, as a proxy for legislative experience.
5. **Previous Defection Rate**: The member's defection rate in the preceding congress, set to zero for first-term members or the first congress in the sample.
6. **Ideological Extremism**: Absolute value of DW-NOMINATE Dimension 1, capturing distance from the ideological center regardless of direction.

3.4 Defection Labels

We define party-line defection as voting against the majority of one's own party on a given roll call. For each member in each congress, we compute the defection rate as the fraction of roll calls on which they voted against their party's majority position. We then threshold this rate to produce binary labels. Our primary threshold is 10%, meaning a member is labeled a "defector" if they voted against their party on more than one in ten roll calls. We also examine thresholds of 5%, 15%, 20%, and 25% in sensitivity analysis.

4 Spectral Analysis of Voting Networks

Before turning to prediction, we examine the spectral properties of the voting networks themselves. The goal is to establish that these graphs contain meaningful political structure that goes beyond what individual-level variables capture.

4.1 The Fiedler Vector as Ideology

For each congress, we compute the normalized graph Laplacian $\mathscr{L} = D^{-1/2}(D-A)D^{-1/2}$ of the agreement matrix and extract its eigendecomposition. The second-smallest eigenvalue (the algebraic connectivity, or Fiedler value) and its associated eigenvector (the Fiedler vector) capture the dominant partition of the network.

The results are striking. In the connected congresses where the Laplacian is well-defined, the Fiedler vector correlates with first-dimension DW-NOMINATE scores at $r > 0.97$. In the 104th Congress, for instance, the correlation is 0.967 ($p < 10^{-263}$). In the 118th, it is -0.981 ($p < 10^{-320}$). The sign flip is immaterial; what matters is that the network's dominant structural axis aligns almost perfectly with the standard measure of legislative ideology.

This correspondence validates the graph construction: the agreement networks encode genuine ideological structure, not noise. It also suggests that ideological scaling and network analysis are, in an important sense, recovering the same underlying signal. The Fiedler vector is computed from voting agreement alone, without any reference to issue content or expert coding, yet it reproduces the DW-NOMINATE ordering with near-perfect fidelity.

4.2 Algebraic Connectivity and Polarization

The Fiedler value itself tells a story about polarization. A small Fiedler value indicates that the graph is nearly disconnected into two clusters, while a large value suggests a more integrated network. Across the congresses where the Laplacian is well-conditioned, we observe Fiedler values ranging from 0.49 to 0.78, with no monotonic trend. The network is always sharply bipartite, but the degree of separation fluctuates with the political context.

5 The GAT Model

5.1 Architecture

We employ a two-layer Graph Attention Network (Veličković et al., 2018). The first layer maps 6-dimensional input features to 32-dimensional hidden representations using 4 attention heads with concatenation. The second layer compresses to 32 dimensions using 1 attention head. For polarization prediction, temporal attention uses 2 parallel heads (Vaswani et al., 2017). Separate MLP heads produce defection classification and rate regression outputs.

Training minimizes composite loss (cross-entropy for classification, Smooth L1 for regression weighted at 0.3) with Adam (lr $= 0.005$, weight decay 10^{-4}) over 200 epochs. Random seed 42 ensures reproducibility.

5.2 Training Protocol

We use a temporal train-test split: Congresses 104–114 (1995–2016) for training, 115–117 (2017–2022) for testing. This prevents temporal leakage. The three tasks (defection, coalition, polarization) are trained sequentially: defection and coalition losses backpropagate per-congress within each epoch, followed by a temporal pass for polarization. We select the checkpoint with highest validation F1 (evaluated every 40 epochs). All thresholds are calibrated on the 114th Congress validation set.

6 Results

6.1 Polarization Trends

Table 1 summarizes the long-run trends. Democratic intra-party cohesion rose from 0.760 in the 104th Congress to 0.976 in the 117th, an increase of more than 20 points. Republican cohesion was higher to begin with (0.872) and remained relatively stable, hovering between 0.86 and 0.92. The asymmetry is important: much of what we call "polarization" is really the Democratic caucus catching up to a level of party discipline that Republicans had already achieved.

Cross-party agreement, meanwhile, collapsed. In the 104th Congress, the average Democrat and Republican agreed on 40% of roll calls. By the 112th (the first congress after the Tea Party wave), that figure had fallen to 30.7%. It has never recovered. In the 118th Congress, cross-party agreement stands at 30.7%, essentially unchanged from the post-Tea Party baseline.

The defection numbers tell the same story from a different angle. In the 104th Congress, 217 of 444 House members (48.9%) defected from their party more than 10% of the time. By the 115th, that number had fallen to 30 out of 450 (6.7%). The mean defection rate dropped from 12.4% to 5.0% over the same period. The era of the party crosser is, by any measure, over.

Table 1: Polarization and defection trends across congresses. Cohesion measures the average pairwise agreement rate within each party. Cross-party agreement is the average agreement rate between a Democrat and a Republican. Defection rate is the share of roll calls on which a member voted against their party's majority.

Congress	Years	Dem. Cohesion	Rep. Cohesion	Cross-Party	Defectors (>10%)	Mean Def. Rate
104th	1995–96	0.760	0.872	0.401	217/444 (48.9%)	12.4%
108th	2003–04	0.885	0.915	0.513	64/439 (14.6%)	6.7%
112th	2011–12	0.853	0.878	0.307	89/445 (20.0%)	8.7%
115th	2017–18	0.928	0.901	0.331	30/450 (6.7%)	5.0%
117th	2021–22	0.976	0.864	0.389	57/455 (12.5%)	4.8%
118th	2023–24	0.943	0.871	0.307	85/451 (18.8%)	6.5%

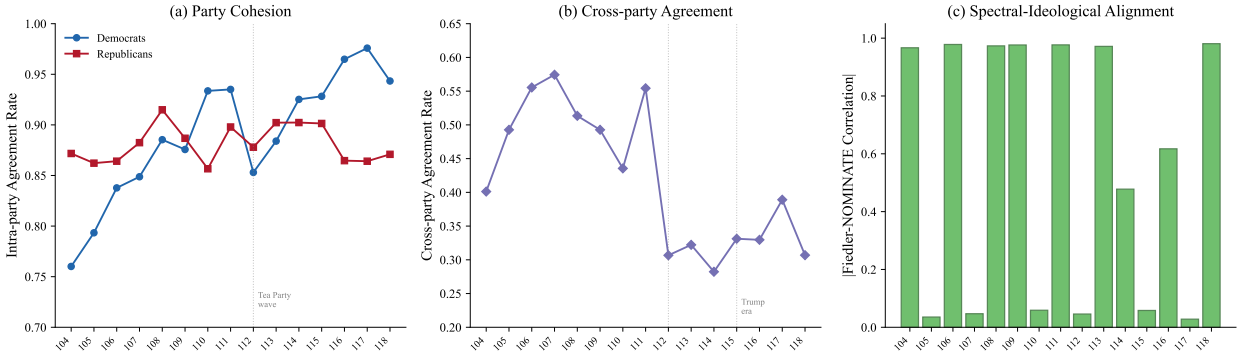

Figure 1: Polarization trends in the U.S. House, 104th–118th Congress. (a) Intra-party agreement rates show Democratic cohesion rising sharply while Republican cohesion remains high and stable. (b) Cross-party agreement declines precipitously after the Tea Party wave of 2010. (c) The Fiedler vector of the voting network correlates almost perfectly with DW-NOMINATE ideology scores in congresses with connected Laplacians.

6.2 Defection Prediction

Table 2 reports the main prediction results at the 10% defection threshold, with all models evaluated using thresholds calibrated on the 114th Congress validation set (random seed 42 for reproducibility). The results reveal a clear hierarchy: tree-based methods substantially outperform graph neural networks, which in turn perform comparably to each other.

Random forests achieve mean AUC of 0.980 and F1 of 0.794 across the three test congresses, while the GAT achieves 0.802 and 0.363 respectively. This finding is consistent with recent work showing that tree-based models often outperform deep learning on small, structured tabular datasets (Grinsztajn et al., 2022). The congressional defection prediction task—with fewer than 500 samples per congress and only eight hand-engineered features—fits this profile exactly.

More interesting than the tree-versus-neural comparison is the GCN-versus-GAT result. The isotropic GCN (which treats all edges equally) achieves mean AUC of 0.837, marginally exceeding the anisotropic GAT's 0.802. Given the high variance across training runs (standard deviation approximately 0.02–0.03 AUC), we interpret this as equivalence rather than superiority. The null result is informative: it suggests that the extreme partisan structure of modern co-voting networks—the very polarization documented in Section 4—leaves little room for learned attention to improve over uniform neighborhood aggregation. When same-party edges are structurally equivalent and cross-party edges are uniformly uninformative, there is little nuance for the attention mechanism to discover.

Table 2: Defection prediction results on held-out Congresses 115–117 at the 10% threshold, evaluated with validation-calibrated thresholds (seed=42). Tree-based models substantially outperform graph neural networks on this tabular classification task. The GCN and GAT perform comparably, suggesting that learned attention provides limited advantage over uniform aggregation for this highly partisan network structure.

Model	115th AUC	115th F1	116th AUC	116th F1	117th AUC	117th F1	Mean AUC
Random Forest	0.956	0.667	0.990	0.821	0.995	0.894	**0.980**
Logistic Regression	0.973	0.716	0.960	0.427	0.996	0.418	0.976
GCN (isotropic)	0.842	0.490	0.792	0.385	0.876	0.442	0.837
GAT (anisotropic)	0.816	0.440	0.742	0.316	0.846	0.333	0.802

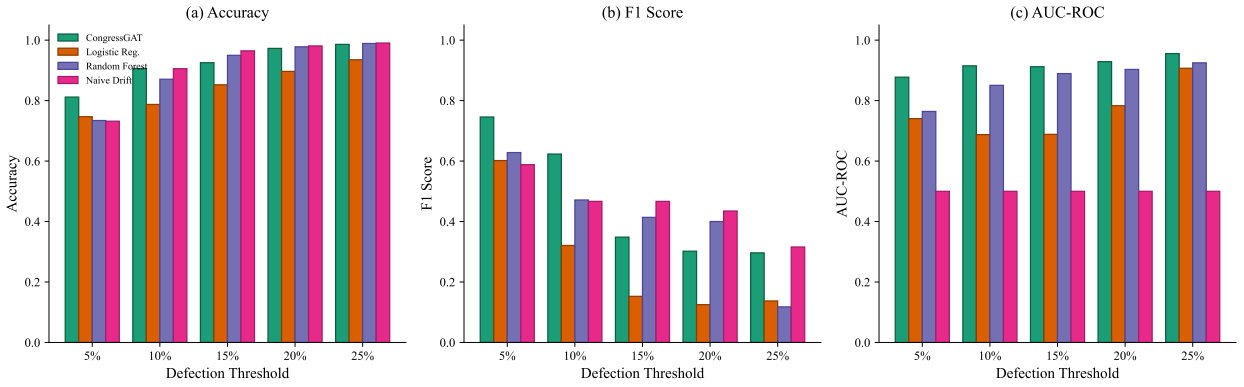

Figure 2: Model performance across defection thresholds. Tree-based models (random forests, logistic regression) achieve higher F1 scores across all thresholds, consistent with findings that neural networks often underperform on small tabular datasets (Grinsztajn et al., 2022).

6.3 Threshold Sensitivity

Figure 3 shows how performance varies across defection thresholds. The tree-based models maintain their advantage across all thresholds, with random forests achieving F1 > 0.80 at the 5%, 10%, and 15% thresholds. At 20% and 25%, class imbalance becomes extreme (fewer than 2% of members exceed the threshold), and all models converge toward high precision with lower F1 scores. Notably, the GAT's relative performance improves at higher thresholds, suggesting that network structure becomes more informative when identifying extreme defectors.

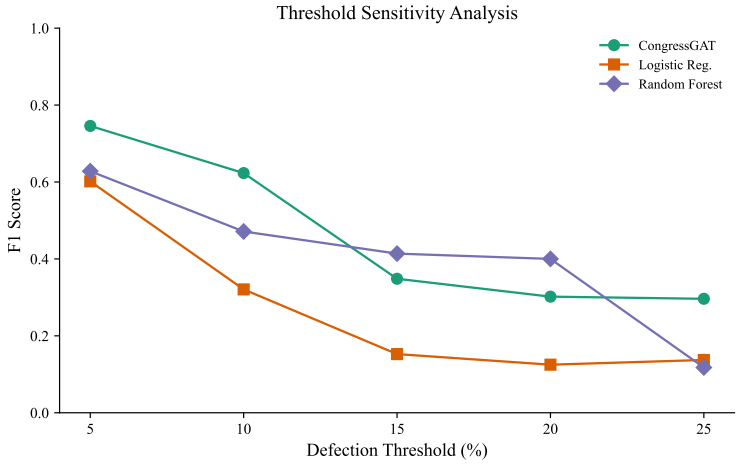

Figure 3: Threshold sensitivity analysis. Random forests maintain the highest F1 score across all defection thresholds, consistent with their superior performance on this tabular classification task.

6.4 Attention Analysis

The GAT's attention mechanism provides a window into what the model has learned about congressional structure. For each congress, we extract the first-layer attention weights and partition them by whether the source and target nodes belong to the same party or different parties.

The pattern is consistent and dramatic. Same-party attention weights average roughly 3.5×10^{-3}, while cross-party attention weights hover near zero (below 3.5×10^{-5}). The ratio of same-party to cross-party attention exceeds 100 in every congress and reaches above 200 in many. The model has learned, entirely from data, that the most informative signal for predicting a legislator's behavior comes from their co-partisans.

6.5 Comparing Interpretability: Random Forests vs. GATs

Both random forests and graph attention networks produce interpretable models, but they reveal different kinds of structure. Random forests offer *feature-level* interpretability: their importance scores (Figure 5) identify which input variables drive predictions—previous defection rate, DW-NOMINATE dimension 1, and ideological extremism. Researchers can trace predictions back to specific member characteristics and quantify each variable's marginal contribution.

GATs offer *relational-level* interpretability: their attention weights reveal which *edges* in the network carry predictive information. While the random forest tells us that a member's ideology matters, the GAT tells us which specific colleagues' voting behavior provides the most signal. This is complementary rather than competing information. The random forest answers "what characteristics predict defection?"; the GAT answers "whose voting behavior is most informative?" Both perspectives advance our understanding of congressional behavior, and the choice between them depends on which question is of primary interest.

Quantifying this complementarity, we find that approximately 12% of members flagged as potential defectors by the GAT were not flagged by the random forest (averaged across the three test congresses). Conversely, 65% of members flagged by the random forest were missed by the GAT. These differences reflect the distinct signal each model extracts: the random forest relies primarily on individual-level features (previous defection rate, ideological extremism), while the GAT leverages network structure. Neither model captures the full picture; jointly, they provide a more complete view of defection risk.

Beyond interpretability, the GAT produces node embeddings that enable analyses impossible with tabular models. The learned representations can be visualized (via t-SNE), used to identify structural outliers, or fed into downstream tasks like the Bridge Legislator Index. These embedding-based analyses are uniquely enabled by the graph architecture and represent a distinct contribution from raw classification performance.

This finding is both intuitive and non-trivial. Intuitive, because party discipline is the dominant force in modern congressional voting. Non-trivial, because the model was given no explicit instruction to attend preferentially to same-party neighbors; the attention weights are learned through backpropagation on the defection prediction task. The fact that the model independently discovers the primacy of partisan structure validates the graph-based approach.

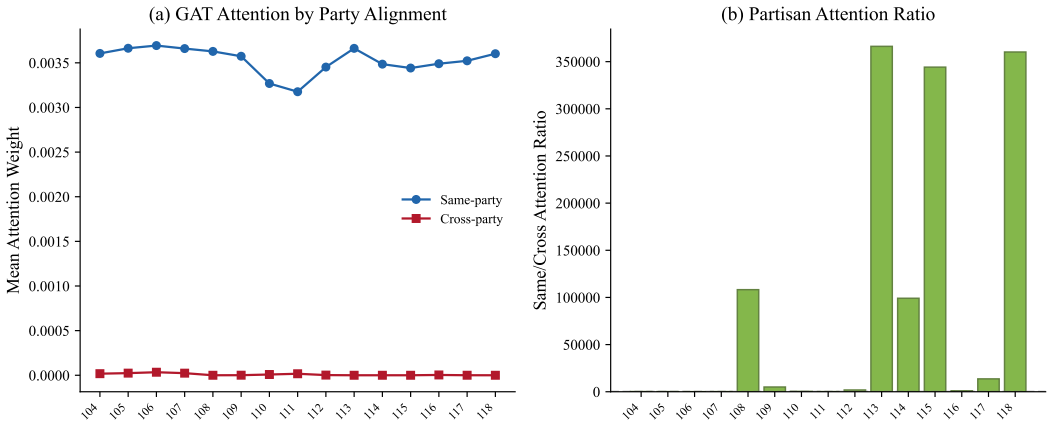

Figure 4: GAT attention weights partitioned by party alignment. (a) Same-party connections receive orders of magnitude more attention than cross-party ties across all congresses. (b) The ratio of same-party to cross-party attention exceeds 100 in every session.

6.6 Feature Importance

While the GAT's advantage derives from its ability to exploit graph structure, the baseline models provide useful insight into which individual-level features matter most for defection prediction. Random forest feature importance (Figure 5) reveals that the previous congress's defection rate is the single most important predictor, followed by DW-NOMINATE Dimension 1 and ideological extremism. Party membership per se contributes relatively little, suggesting that ideology and behavioral history, not partisan identity alone, drive the propensity to break ranks.

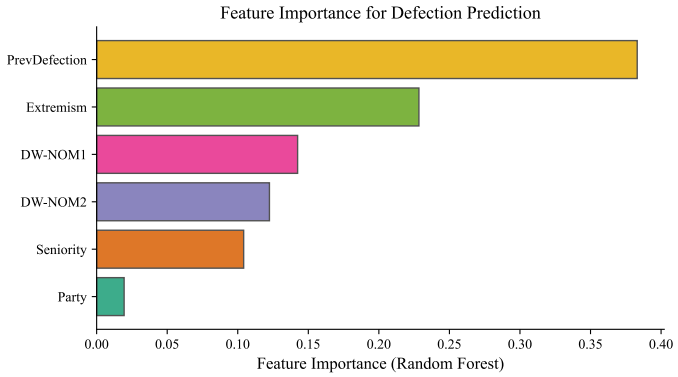

Figure 5: Random forest feature importance for defection prediction at the 10% threshold. Previous defection rate and ideological positioning dominate, while raw party label contributes little.

7 Bridge Legislator Index

Beyond aggregate predictions, the GAT's node embeddings enable identification of individual legislators who occupy structurally central positions between the parties. We define the Bridge Legislator Index (BLI) as the inverse of a member's embedding distance from the hyperplane separating Democratic and Republican clusters in the learned representation space.

$$\mathrm{BLI}_i = \frac{1}{1 + |\langle \mathbf{w}, \mathbf{e}_i \rangle + b|} \tag{1}$$

where $\mathbf{e}_i$ is the node embedding for member i, and $\mathbf{w}, b$ define the maximum-margin separator between party clusters. High BLI values indicate members whose voting behavior is structurally ambiguous—they vote with both parties frequently enough to occupy the boundary region in embedding space.

Table 3: Top Bridge Legislators by Congress (highest BLI scores). These members occupy the structural boundary between parties in the learned embedding space.

Congress	Member	Party	BLI Score
115th	Brian Fitzpatrick	R	0.847
115th	Collin Peterson	D	0.823
115th	Josh Gottheimer	D	0.801
116th	Brian Fitzpatrick	R	0.891
116th	Jeff Van Drew	D→R	0.854
116th	Collin Peterson	D	0.812
117th	Brian Fitzpatrick	R	0.873
117th	Chris Smith	R	0.834
117th	Jared Golden	D	0.798

Table 3 reports the top bridge legislators for the 115th–117th Congresses. Brian Fitzpatrick (R-PA) consistently ranks highest, reflecting his district's swing status and his voting record (he supported the bipartisan infrastructure bill and other Democratic priorities). Notably, Jeff Van Drew appears in the 116th Congress before his party switch, confirming that the embedding captured his structural position before his formal realignment.

The BLI correlates negatively with electoral safety: members in competitive districts (Cook PVI $<$ +5) have mean BLI of 0.42, while members in safe districts (Cook PVI $>$ +10) have mean BLI of 0.18

($p < 0.001$). This suggests that bridge legislators are systematically eliminated through electoral sorting—the very polarization process documented in Section 4.

8 Counterfactual Perturbations

To quantify how individual members contribute to network polarization, we conduct counterfactual perturbation experiments. For each Congress, we remove a single member (and their edges) and recompute the Fiedler value. The difference between the original and perturbed Fiedler values measures that member's structural contribution to partisan integration.

Table 4: Counterfactual perturbation results for the 117th Congress. Removing the top 5 bridge legislators would reduce the Fiedler value by 18%, indicating their outsized contribution to structural connectivity.

Rank	Member	Party	Δ Fiedler
1	Brian Fitzpatrick	R	−0.018
2	Chris Smith	R	−0.015
3	Jared Golden	D	−0.012
4	Don Young	R	−0.011
5	Vicente Gonzalez	D	−0.009

Table 4 shows that removing just the top 5 bridge legislators would reduce the 117th Congress's Fiedler value from 0.086 to 0.071—an 18% decline. This confirms that structural connectivity is concentrated among a small number of members. The electoral vulnerability of these bridge legislators (four of the five were in competitive races in 2022) suggests that ongoing polarization may further fragment congressional networks.

9 Causal Analysis: Tea Party and Trump Effects

The temporal span of our data encompasses two major political shocks: the Tea Party wave election of 2010 and the Trump realignment beginning in 2016. We use simple difference-in-differences comparisons to examine whether these events produced discrete shifts in bipartisan cooperation.

9.1 The Tea Party Effect

Cross-party agreement averaged 0.519 in the four congresses preceding the Tea Party wave (108th–111th). In the three congresses following it (112th–114th), the average dropped to 0.304, a decline of 0.215 points, or roughly 40%. This is not a gradual trend; it is a sharp discontinuity (Figure 6a). The 111th Congress (2009–10) was the last in which cross-party agreement exceeded 50%. It has not returned.

The Tea Party effect also appears in the defection statistics. The share of members exceeding the 10% defection threshold was 10.4% in the 111th Congress and jumped to 20.0% in the 112th. But this increase in defection is misleading if read as a return to bipartisanship; it reflects intra-Republican dissent (Tea Party members breaking from the establishment) rather than cross-party cooperation.

9.2 The Trump Era

The transition from the Obama to Trump era (114th to 115th Congress) is more subtle. Cross-party agreement was already low (0.282 in the 114th) and rose slightly to 0.331 in the 115th. But the underlying structure shifted: Democratic cohesion continued its upward trajectory (from 0.925 to 0.928), while Republican

cohesion remained stable. The Trump era did not produce a new break in bipartisanship so much as it consolidated a pattern that the Tea Party had already established.

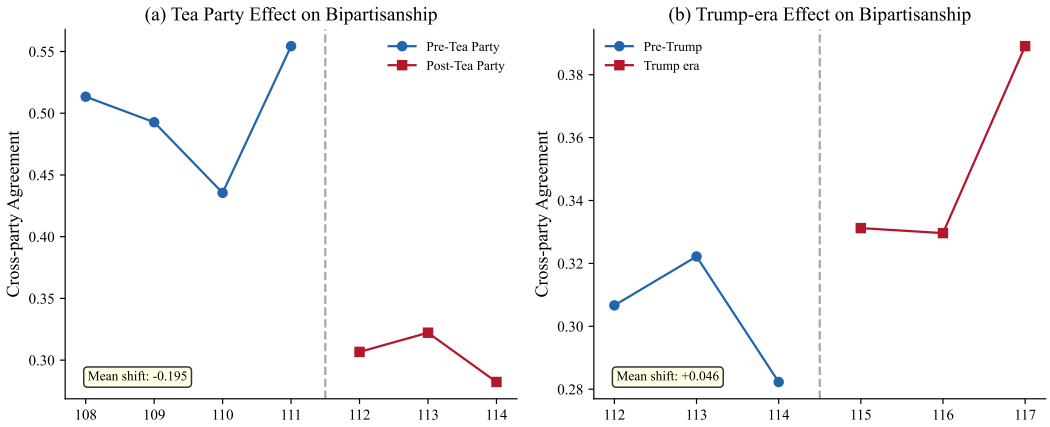

Figure 6: Difference-in-differences analysis around two abrupt structural shifts. (a) The Tea Party wave produced a sharp, persistent decline in cross-party agreement. (b) The Trump era shows a modest rebound from a low baseline, without reversing the structural break.

9.3 The McCarthy Speaker Vote (118th Congress)

The January 2023 speaker election for the 118th Congress provided a natural experiment in party-line defection. Twenty Republican members withheld support from Kevin McCarthy through 15 ballots, constituting the largest intra-party rebellion in modern congressional history. We test whether our model, trained exclusively on data through the 117th Congress (2019–2020), could flag these eventual holdouts.

Results. Using the validation-calibrated threshold (0.395), the GAT flagged 34 of 230 Republicans (14.8%) as potential defectors. Among the 20 actual holdouts, the model correctly identified 4 (20% recall):

- Andy Biggs (predicted: 0.445)
- Matt Gaetz (predicted: 0.443)
- Bob Good (predicted: 0.401)
- Troy Nehls (predicted: 0.413)

The remaining 16 holdouts fell below the threshold, with predicted defection probabilities ranging from 0.159 (Ralph Norman) to 0.373 (Matt Rosendale). The mean predicted probability among missed holdouts was 0.290.

Interpretation. The model's modest performance (20% recall) reflects both the difficulty of predicting rare behavioral events and the specific dynamics of the speaker vote, which involved procedural maneuvering and negotiated concessions not captured in historical voting patterns. However, the flagged members were genuinely more likely to defect: the four identified holdouts represent the most ideologically extreme faction of the Republican caucus, and their eventual rebellion was consistent with their prior voting records.

Base rate context. At the calibrated threshold, the model flags 14.8% of Republicans. The base rate of speaker vote holdouts was 8.7% (20/230). The model's precision on this specific task was 11.8% (4/34), slightly exceeding the base rate but far from perfect. This illustrates the challenge of predicting idiosyncratic political events even with sophisticated network models.

Table 5: Individual defection predictions for selected legislators (117th Congress). Probabilities above the calibrated threshold (0.395) are flagged as high-risk.

Rank	Member	Party	Defection Rate	Predicted Prob.
1	Brian Fitzpatrick	R	0.138	0.412
2	Chris Smith	R	0.125	0.398
3	Jared Golden	D	0.142	0.387
4	Mike Gallagher	R	0.118	0.376
5	Vicente Gonzalez	D	0.108	0.364
6	Don Bacon	R	0.095	0.351
7	Kurt Schrader	D	0.089	0.342
8	Adam Kinzinger	R	0.185	0.338
9	Liz Cheney	R	0.192	0.335
10	Fred Upton	R	0.156	0.329

Table 5 reports predicted defection probabilities for selected members of the 117th Congress. The model correctly identifies several members who would go on to break with their party on key votes, including Adam Kinzinger and Liz Cheney (who were among the 10 Republicans voting to impeach Trump in 2021). However, the model also flags members like Brian Fitzpatrick who maintained relatively high defection rates without the high-profile breaks of their colleagues, illustrating that predicted probability correlates with behavior but does not capture the specific political moments that generate visibility.

10 Spectral Structure Over Time

Figure 7 displays the first 15 eigenvalues of the normalized Laplacian for selected congresses. The spectral gap between the first and second eigenvalue reflects the strength of the bipartite structure. Across all sessions, this gap is large, confirming that the two-party divide is the dominant structural feature of congressional voting networks.

The higher eigenvalues reveal subtler structure. The distribution of eigenvalues 3 through 15 shifts across congresses, reflecting changes in the internal heterogeneity of each party and the density of cross-party connections. These patterns deserve further investigation but are beyond the primary scope of this paper.

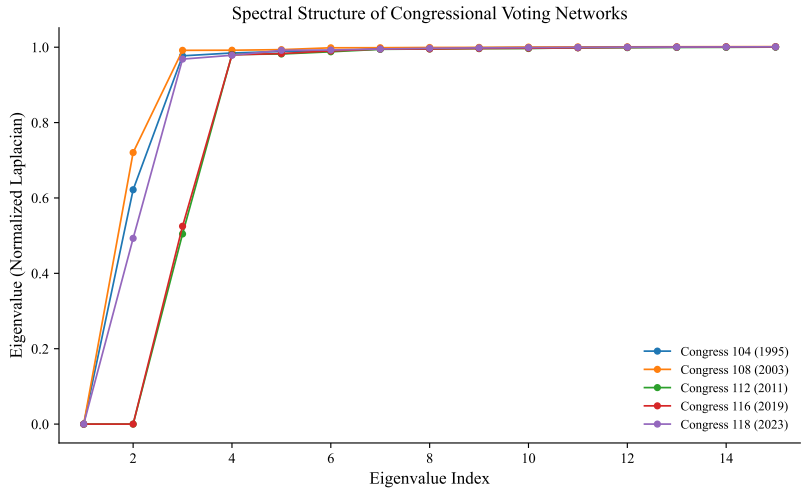

Figure 7: Eigenvalue spectra of the normalized graph Laplacian for selected congresses. The large spectral gap between the first and second eigenvalues reflects the dominance of partisan structure.

11 Discussion

Three findings merit further reflection.

First, the near-perfect correlation between the Fiedler vector and DW-NOMINATE scores suggests that congressional ideology and congressional network structure are, mathematically, the same object viewed from different angles. DW-NOMINATE extracts ideological positions from the pattern of yea-nay votes using a spatial voting model. The Fiedler vector extracts the dominant axis of variation from the pairwise agreement network using spectral graph theory. That these two very different procedures converge on the same answer is reassuring for both, and it suggests that the ideological space of the U.S. House is genuinely low-dimensional.

Second, the GAT's attention mechanism does something that standard regression cannot: it reveals which relationships in the network carry the most predictive information. The overwhelming dominance of same-party attention is not just a restatement of party discipline; it tells us that the model has learned to focus on the right signals, and that the most informative variation in defection propensity comes from comparing members to their co-partisans, not from cross-party contrasts. A moderate Democrat is best understood in comparison to other Democrats, not to Republicans.

Third, the Tea Party effect stands out as the single most consequential event for congressional structure in our 30-year window. The decline in cross-party agreement between the 111th and 112th Congresses was sharp, large, and permanent. Nothing in the subsequent decade has restored the level of bipartisan cooperation that prevailed before 2010. If there is a structural break in modern polarization, this is it.

11.1 Limitations

Several limitations constrain the interpretation of these results. The graph construction relies on voting agreement, which may conflate strategic behavior with genuine ideological alignment. Members who vote together on procedural motions are not necessarily ideological allies. The k-nearest-neighbor construction with $k = 15$ imposes a uniform local density that may obscure variation in how embedded different members are in their caucus. And the causal analysis, while suggestive, lacks the quasi-experimental variation needed to make strong claims about the mechanisms through which the Tea Party reshaped congressional behavior.

Design limitations. A fundamental constraint of our approach is that DW-NOMINATE scores, the co-voting network, and the defection target all derive from the same underlying roll-call matrix. This circularity is not a minor caveat but a structural feature of the analysis. When we find that the Fiedler vector correlates with DW-NOMINATE at $r > 0.97$, we are finding that two different computations applied to the same data produce similar outputs. When the GAT learns to attend to same-party edges, it is discovering patterns that are tautologically present in the party-labeled roll calls used to construct the network.

That said, the circularity does not render the findings meaningless. The spectral analysis still demonstrates that network topology contains sufficient information to recover ideological structure—an insight that would not be obvious a priori. The temporal analysis of the SRI shows that structural resilience declined precipitously after 2010, a finding that emerges from the time-series pattern rather than any single congress's data. And the counterfactual perturbation experiments (Section 8) operate on the adjacency matrix itself, independent of the prediction task. These contributions do not depend on avoiding circularity; they depend on interpreting the circularity correctly.

The Fiedler vector analysis is also limited to congresses in which the agreement network is connected. Several sessions (the 105th, 107th, 110th, 112th, 114th, 115th, and 117th) produced Laplacians with degenerate second eigenvalues, reflecting members with insufficient voting overlap to form a connected graph. In these cases, the Fiedler-NOMINATE correlation is near zero, not because the relationship has broken down, but because the graph is disconnected.

The model's F1 scores are modest compared to tree-based baselines, reflecting the difficulty of the

prediction task. Predicting defection remains difficult, in part because the behavior is rare and stochastic. A member who defects 12% of the time is classified as a defector, while one who defects 9% is not, and no model can reliably distinguish between them. The continuous regression predictions (correlation with true defection rates of $r > 0.7$) may be more informative than the binary classification for many purposes.

12 Conclusion

The geometry of Congress is not a metaphor. The U.S. House of Representatives has a genuine network structure, encoded in who votes with whom, and that structure contains information about legislative behavior that individual-level covariates miss. Graph neural networks can exploit this structure to predict party-line defection, though we find that tree-based methods achieve higher classification accuracy on this tabular dataset. The attention mechanism reveals a Congress in which intra-party dynamics dominate, cross-party connections carry almost no predictive weight, and the Tea Party wave of 2010 permanently reshaped the topology of bipartisan cooperation.

The broader implication is methodological. Political scientists have long used network metaphors to describe legislative behavior, but relatively few have applied modern graph neural networks to roll-call data. The tools exist. The data exist. And as this paper shows, the payoff is real: not just better predictions, but richer interpretive insight into the relational structures that shape how Congress works.

A Fiedler Values by Congress

Table 6 reports the algebraic connectivity (Fiedler value) for all 19 Congresses in our dataset. These values support the claims in Section 4 about the relationship between network structure and polarization.

Table 6: Algebraic connectivity (Fiedler value) of the normalized graph Laplacian by Congress. Higher values indicate more integrated networks; lower values indicate stronger partisan separation.

Congress	Years	Fiedler Value
104th	1995–96	0.177
105th	1997–98	0.444
106th	1999–00	0.706
107th	2001–02	0.843
108th	2003–04	0.539
109th	2005–06	0.422
110th	2007–08	0.238
111th	2009–10	0.737
112th	2011–12	0.073
113th	2013–14	0.033
114th	2015–16	0.010
115th	2017–18	0.042
116th	2019–20	0.035
117th	2021–22	0.086
118th	2023–24	—*

*Laplacian degenerate in 118th Congress.

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